

# STAT 672 Time Series Course Project

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## STAT672 Syllabus: Requirements for Project Report

Project: The project is aimed at fostering students to apply the statistical methods to solve a real-world problem of time series. Each student works individually on a project pre-approved by the instructor.

Student shall submit a project proposal that describes the **topic**, **background** and **objective** of your project, **problem description**, and **data generation** by Jan 25.

Student is required to give a (15 - 20 minutes) **presentation** in class in the end of semester and submit **a project report** which includes the topic, background and objective of your project, problem description, executive summary, data generation, materials and methods, results, conclusions and appendices.

The project report is due May 3. The grade will be based on the complexity of the real problem and quality of project.

## STAT672 Syllabus: Course Objectives:

This course introduces fundamental concepts and statistical models for time series data which could be modeled by Gaussian distribution. Topics include **stationarity**, **autocorrelation** and **partial autocorrelation**; **ARIMA models**; **ARCH** and **GARCH** models; **model estimation and forecasting**; using R for the analysis of time series; and applications of time series.

Title for course project: Time Series Forecasting for Electricity Usage

## Background:

To address climate change, on one hand we can discovery catalyst to discovery new energy resources, on the other hand we also should be frugal on what we already have. Thus, to predict the electric power distribution precisely on demand can save energy.

The electric power distribution problem is the distribution of electricity to different areas depends on its sequential usage. But predicting the following demand of a specific area is difficult, as it varies with weekdays, holidays, seasons, weather, temperatures, etc. However, no existing method can perform a long-term prediction based on super long-term real-world data with high precision. Currently, without an efficient method to predict future electric usage, managers have to make decisions based on the empirical number, which is much higher than the real-world demands. It causes unnecessary waste of electric and equipment depreciation.

On the other hand, the oil temperatures can reflect the condition of electricity Transformer. One of the most efficient strategies is to predict how the electrical transformers' oil temperature, which is safe and can avoid unnecessary waste.

## Objective:

The primary objective of this course project is to evaluate different time series models that covered in this course and also to gain better understanding on model selection by considering the topics of Linear statistical time-series models vs. non-linear time series, and Short-term vs. long-term.

## Data:

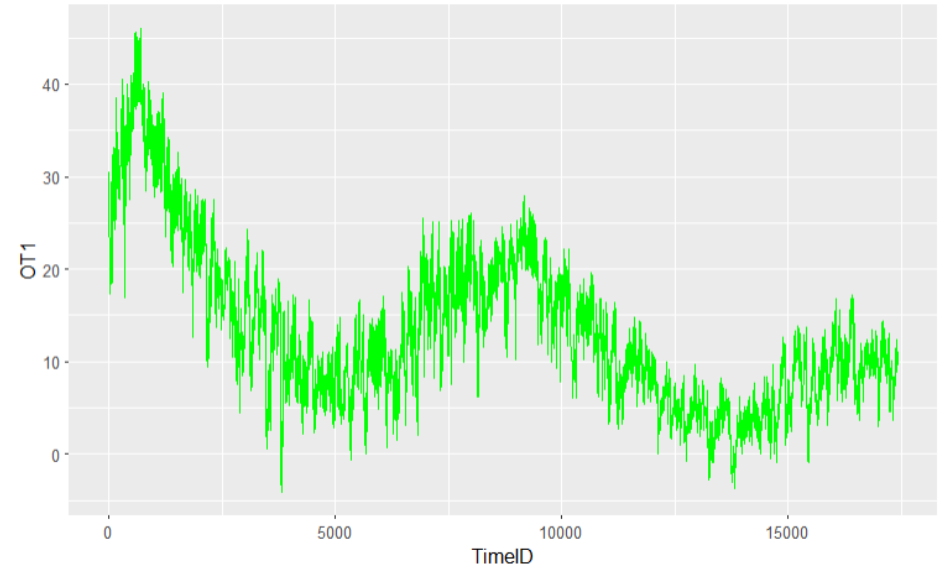
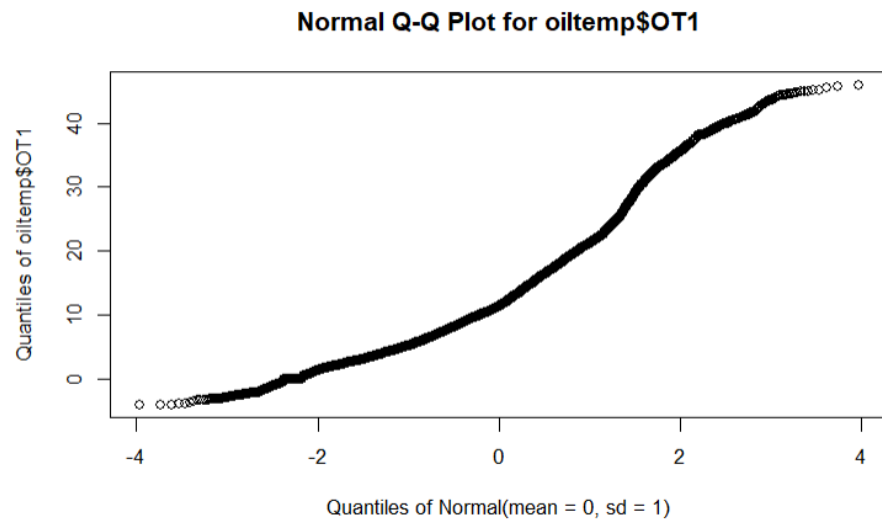
### Long-term time series

The data set is oil temperatures at 2 stations, total 17420 time points in hourly scale. The oil temperature at each station relates to 5 variables and one data column shown as in the table below.

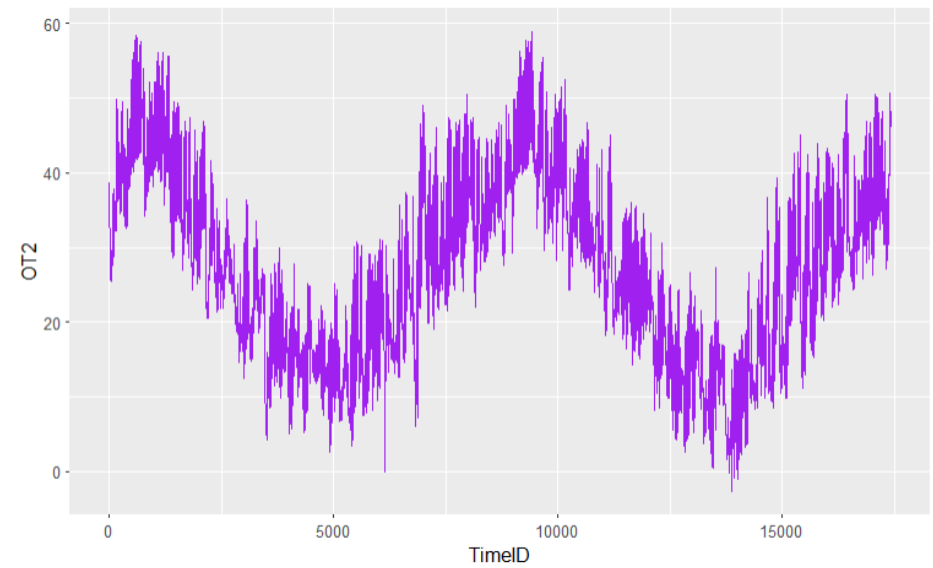
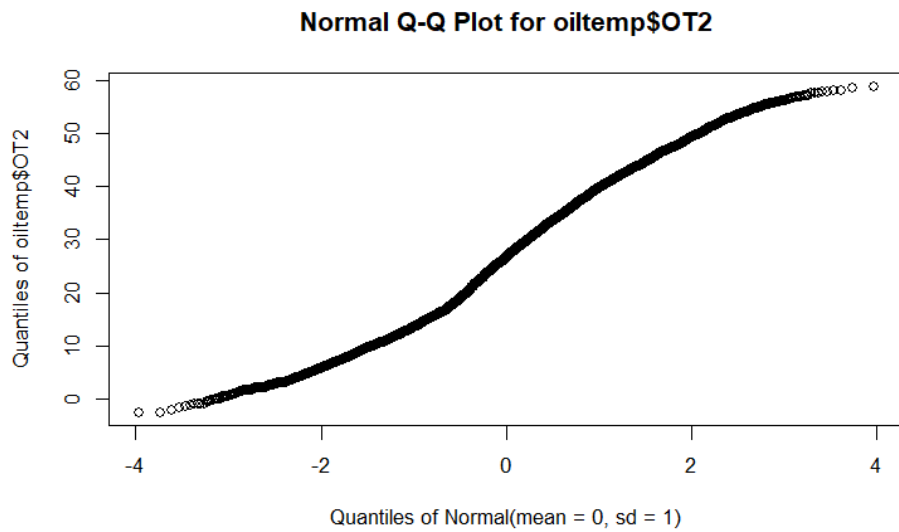
| Date                        | HUFL             | HULL              | MUFL               | MULL                | LUFL            | LULL             | OT                                    |
|-----------------------------|------------------|-------------------|--------------------|---------------------|-----------------|------------------|---------------------------------------|
| The recorded data in hourly | High useful load | High useless load | Middle useful load | Middle useless load | Low useful load | Low useless load | Oil Temperature (the target variable) |

# Gaussian Distribution

Target variable: Oil Temperature at Station 1

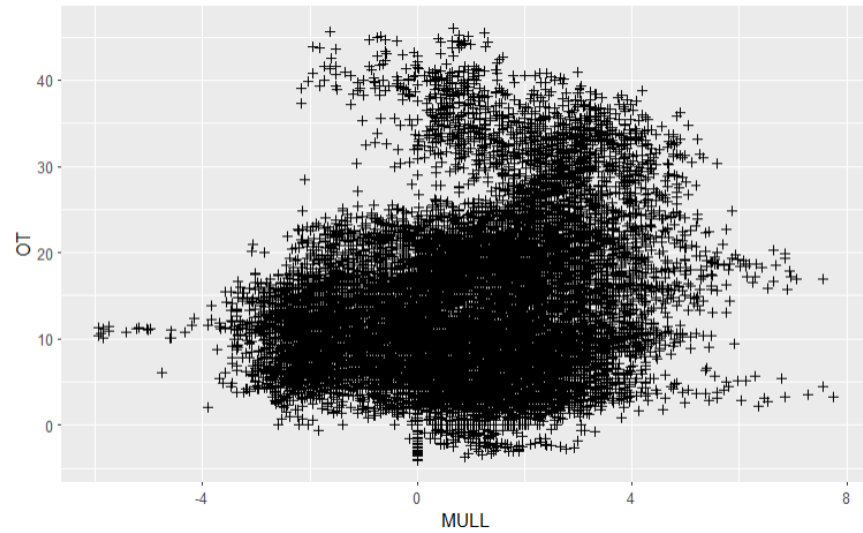
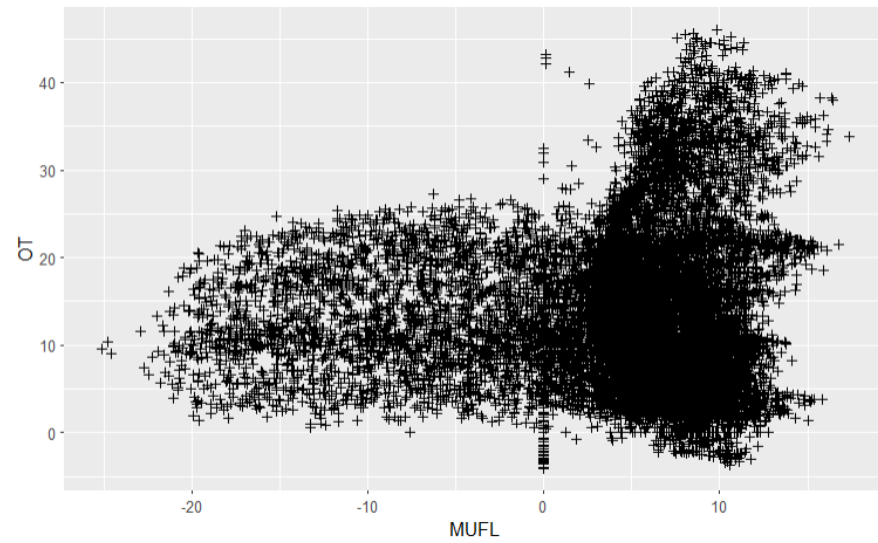
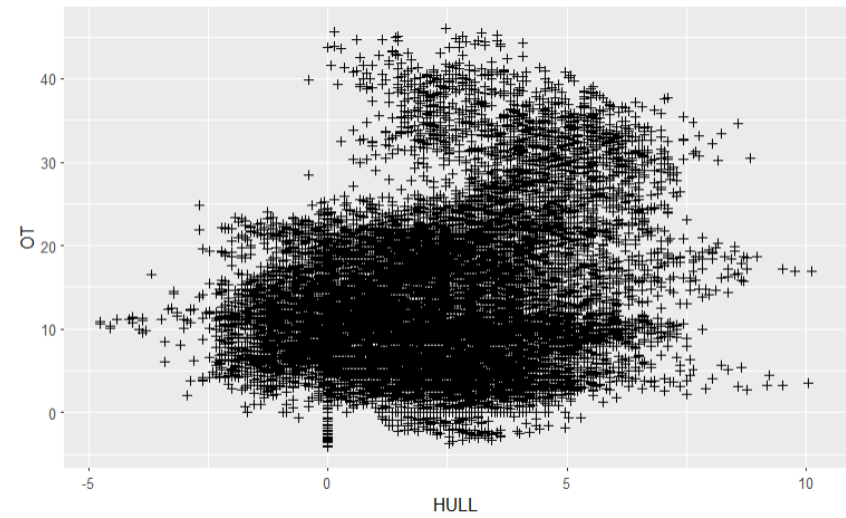
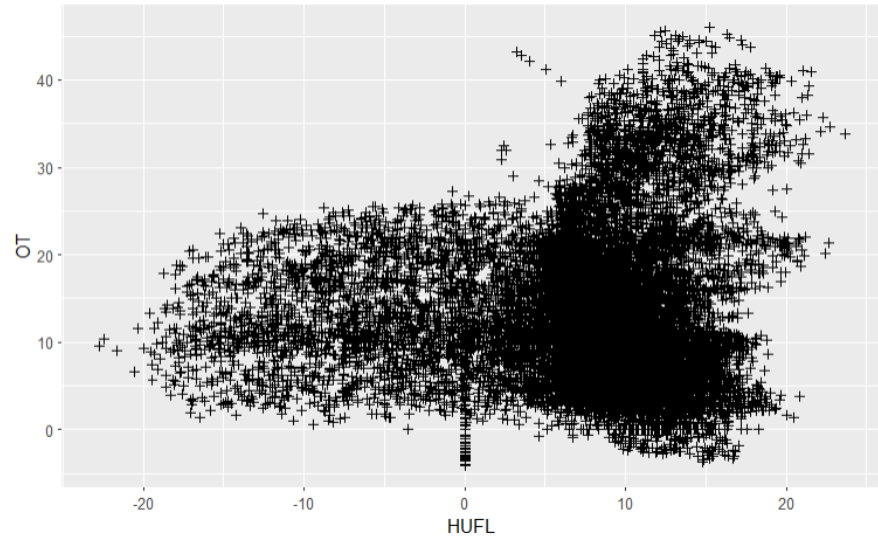


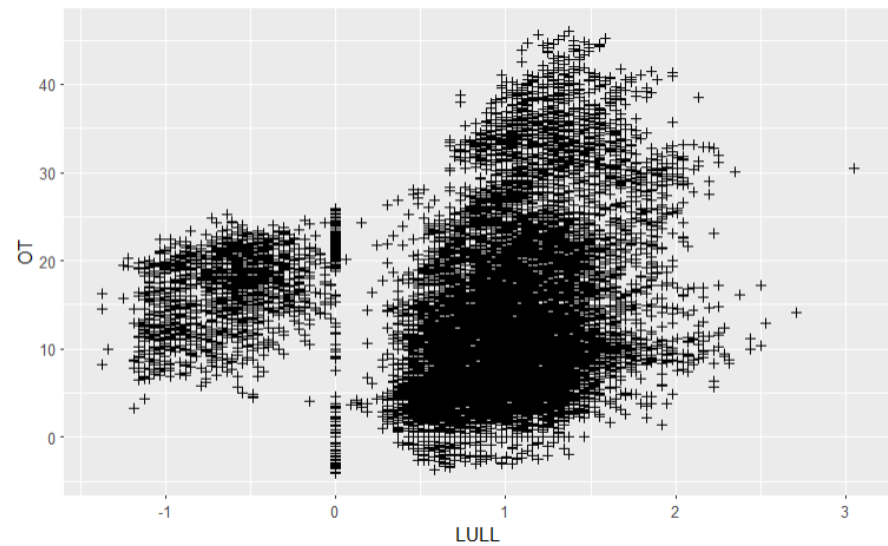
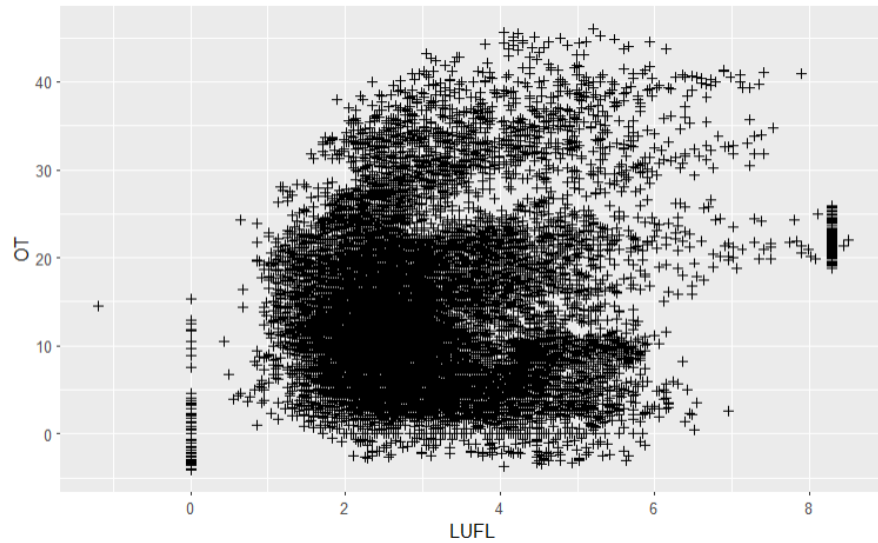
Target Variable: Oil Temperature at Station 2



## Scatter Plot

Variables versus Target Variable: Data at Station 1 "HUFL" "HULL" "MUFL" "MULL" "LUFL" "LULL" versus "OT"





## Data usage

There are 3 csv files: the variables and oil temperature in station 1, the variables and oil temperature 2, and the file contains only the two oil temperatures series. This dataset enables us to explore in multiple ways, forecasting oil temperature in only one station considering other variables in linear statistical time series models. The file contains two oil temperatures can be used as a multivariate time series, oil temperature series at each station regarded as one variable.

## Computational Task

- 1- Data Analysis and decomposition,
- 2- Forecasting oil temperature by statistical linear time series model or non-linear model by considering the variables in the model,
- 3- Gain better understanding on model selection by considering the topics of Linear statistical time-series models vs. non-linear time series and Short-term vs. long-term through this project.

## Reference

[1] Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting, Haoyi Zhou, et al. 2021.